Fieldnote

Source-linked operational answers, with shared facts and private conversational memory.

Browser chat → Node.js API → Gemma plan + scope checks → deterministic tools → SQLite / FTS5 + sources → tables, charts and citations

Data and retrieval

The released snapshot contains 600 wells, 1,25,198 inspections and 17,415 maintenance records. Node.js 24 and SQLite/FTS5 retain original JSON/JSONL records, linked free-text reports, manuals, SOPs and new letters/MOC/recall documents. Newer releases supersede changed record IDs while retaining earlier versions. Type and team history, late-entered dates, cost revisions and structured corrections are applied.

Agentic layer and deterministic tools

Gemma 4 proposes a schema-constrained query plan. Literal scope, dates and operations are checked; numerical results come from linked data tools. Tools handle inspection status, gas-response deadlines, bypass timelines and memo extensions, laboratory evidence, current pump-lot inventory, cold-weather checks, reported delays and temperature comparisons. Unknown reservoir pressure, oil output and incident-specific H2S readings are not invented. Risk ranking is explicitly a heuristic judgement grounded in operator quotes and warning signs.

Memory, feedback and provenance

Each user has persistent private conversations, topic/entity sets and display preferences. “First question” returns to that session's first result set even after unrelated topics. Shared replacement reports retain the speaker, effective date, affected installation and verification state. Direct responsible-user reports are distinguished from hearsay; missing work orders remain visible. Preferences such as sorting and currency units apply only to their author. Corrections preserve originals and recompute subsequent answers.

Interface and evidence

Questions and answers are shown in a compact chat interface with cards, tables, charts and maps. Sources, complete paginated data and technical traces open on demand. Every final turn was typed through the actual browser UI under the named user/session; full response JSON and screenshots were saved. Unsupported or malformed model output is rejected, with a labeled local fallback where supported.

Models, costs and boundaries

Model: gemma-4-26b-a4b-it via Google's Gemini API. This final browser replay had 18 model-backed read turns, 3 local memory/feedback writes, and 0 provider fallbacks. Provider-reported usage: 16,392 input and 311 output tokens; failed requests may lack usage. The supplied guide describes Gemma as free-tier; invoice cost was not measured. The result is a hybrid application result, not unguarded model accuracy.

Next: broader paraphrase evaluation, stronger semantic checks for novel SQL, authenticated deployment and richer event-time memory. AI-assisted development using OpenCode. The accompanying automated demo uses labeled synthetic narration. Application: MIT. Dataset and fictional documents: ©2026 FlytBase, MIT.

Evidence and operational decisions

Q1 · Reservoir pressure

Unavailable: there are no reservoir-pressure measurements. Equipment pressure ratings and flowline tests cannot support a reservoir-pressure map. Output: explicit missing-data answer; no invented map.

Q2 · Best oil producer in 2026

Unavailable: no metered oil-production volumes or rates. Temperature indicates operating state, not oil volume; spills are not production. Output: no unsupported well ranking.

Q3 · Gas-response deadlines

26 reports: **17 met; 9 missed.** OPS-1 uses 24 hours before January and 12 hours from 1 Jan 2027. A later maintenance start or manual inspection on the same well qualifies; a drone inspection does not. Selected output rows:

Well	Reported (UTC)	Response	Hours	Limit	Result
WELL-0149	2026-11-06T18:20:00Z	MNT-0149-029	13.67	24	Met
WELL-0163	2026-11-03T20:55:00Z	MNT-0163-019	251.08	24	Missed
WELL-0324	2027-01-01T22:40:00Z	MNT-0324-036	9.33	12	Met

Q4 · Temporary bypasses

3 currently in place; 6 current/historical exception rows. The MOC-031 extension in MOC-044 is applied. Complete six-row status output:

Well	Installed	Removed	Approval	Valid until	Status
WELL-0158	2026-11-20	2026-12-28	MOC-2026-036	2026-12-15	Expired
WELL-0405	2026-12-20	still in	none	—	Not approved
WELL-0248	2026-11-05	still in	MOC-2026-044	2027-02-28	Allowed
WELL-0249	2026-11-25	still in	MOC-2026-036	2026-12-15	Expired
WELL-0561	2026-12-08	2026-12-19	none	—	Not approved
WELL-0087	2026-12-02	2027-01-06	MOC-2026-036	2026-12-15	Expired

Q5 · Regulator letter

6 covered spills; 3 noncompliant. The letter requires the laboratory-sample sentence in the clean-up section; field soil readings do not substitute. All six output decisions:

Well	Inspection	bbl	Clean-up job	Compliant?
WELL-0119	INS-0119-0209	6	MNT-0119-029	Yes
WELL-0123	INS-0123-0093	6	MNT-0123-014	No: lab sentence absent
WELL-0168	INS-0168-0190	5.6	MNT-0168-024	Yes
WELL-0182	INS-0182-0208	5.2	MNT-0182-028	No: lab sentence absent
WELL-0262	INS-0262-0218	6.1	MNT-0262-029	No: lab sentence absent
WELL-0277	INS-0277-0205	6.2	MNT-0277-031	Yes

Q6 · Conflicting current status

15 disagreements. Compare the latest eligible inspection and explicit status-changing note after corrections. Newer shut-ins take precedence; newer restarts remain unconfirmed. Selected output decisions:

Well	Area	Current state
WELL-0010	Southwest Block 23	Not producing oil
WELL-0026	Southwest Block 23	Producing oil (unconfirmed)
WELL-0032	Southwest Block 23	Producing oil (unconfirmed)

Output excerpts keep this report within three pages. Complete answers, every result row and visuals · 21 browser screenshots · Source repository. Recorded replay: 21 turns, 18 model reads, 3 local writes, 0 fallbacks. These checks are not an official answer key.

Memory, priorities and comparisons

Q7 · Repair triage (both turns)

10 candidates with recent warning notes and no subsequent repair. First three crew priorities are below. Ranking uses safety, drive trips, thermal decline, nonproduction and corrective history; it is a heuristic, not a calibrated 90-day failure probability.

Rank	Well	Operator evidence (excerpt)
1	WELL-0219	2027-01-05T09:55:00Z — Kayla Morgan: “WELL-0219 drive tripping on underload, restarted from the panel.” 2027-01-13T06:20:00Z — Marcus Lee: “Drive on WELL-0…
2	WELL-0332	2026-12-31T19:40:00Z — Brandon Shaw: “Rod squeal at WELL-0332 stuffing box, greased the polished rod.” 2027-01-03T13:15:00Z — Austin Kemp: “Pumper reports …
3	WELL-0336	2026-12-24T09:00:00Z — Brandon Shaw: “WELL-0336 amps hunting on the drive panel, still running.” 2027-01-07T15:05:00Z — Austin Kemp: “WELL-0336 drive tripp…

Q8 · Attribution and privacy (five turns)

Nathan’s direct WELL-0333 replacement report is shared, effective yesterday, with paperwork pending. His area/ID sorting and costs-in-thousands preference stays private. Robert sees **8 documented recalled pumps**. Olivia’s WELL-0577 hearsay remains unconfirmed and does not remove it. Nathan sees **3** in his team, with his preferred units:

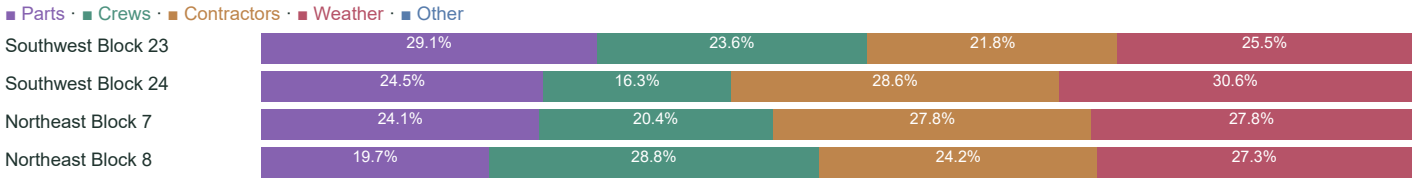
Well	Area	Installation ($000)
WELL-0305	Northwest Block 12	24.528
WELL-0426	Northwest Block 12	24.044
WELL-0428	Northwest Block 12	27.074

Q9 · Topic memory (six turns)

The first recalled-pump set survives topic changes. Nathan Pierce leads **150 Northwest wells**. Cold-weather rule: producing at ≤ −5°C during Dec–Feb requires a heat-trace job within 7 days; missed example: **WELL-0310** (INS-0310-0213). WELL-0332’s H2S reading is **not recorded**. Returning to the original pump set yields **2 nonproducing wells**: WELL-0160, WELL-0263. Final priority: **WELL-0332**, with unresolved gas-response risk ahead of equipment concerns.

Q10 · Extra shut-in waiting

474 jobs explicitly quantify extra waiting; 709 other corrective jobs lack that statement and are not zero-delay observations. Shares use documented delayed jobs within each area; “typical” is median extra days. **Southeast Block 31: highest area median, 10 days.** Stacked output excerpt (first four areas; full area table/chart linked below):



Q11 · Consistently hotter?

For 2026 producing readings, each well is weighted equally within type/month. Southeast leads raw wellhead temperatures. After ambient adjustment and equal weighting of **36 common type/month strata**, Southwest leads the mean but wins only **17/36**: no universally hottest region. This does not measure reservoir heat or oil output.

Region	Raw °C	Matched Δ°C	Wins / 36
Southwest	50.21	31.19	17
Northeast	50.19	31.16	11
Southeast	52.75	31.02	6
Northwest	51.62	30.9	2

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